

Alpilla 600 MW Combined Cycle Power Plant

EIA Conditions Compliance Matrix

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A	IFR	2026-12-14	First issue for Owner review (open item TQ-IEC-026)	EPCE	EPC HSE lead	EPC engineering manager

1. Purpose and scope

1.1 This matrix lists every binding condition of the positive EIA decision (Owner document ALP-OWN-ENV-001 Rev 0, source SRC-OWNR-0007) and shows for each condition how the design and the construction of the Works comply, and which project document is the evidence. It implements requirement ER-14.01.

1.2 The matrix covers the conditions on air emissions, water, noise, marine works and ecology and other construction conditions, and the waste requirements of the Environmental Law (No. 2872) that ER-14.01 adds. One row is provided per condition; a condition that is not yet met or depends on a document not yet issued is marked 'preliminary' in the status column of the record and is listed in section 5.

1.3 Compliance values are not repeated in this matrix: each row cites the design parameter, guarantee, decision or permit record that holds the value. The EIA limit quoted in the condition column is the Owner's text and governs where it differs from a design value.

1.4 The matrix is a living document: it is re-issued with the evidence documents, and at the latest before the Tests on Completion, so that the acceptance criterion of ER-14.01 (no outstanding EIA non-conformities at Taking-Over) can be demonstrated.

2. Governing requirements and design data

ER-14.01: The Works shall comply with the conditions of the positive EIA decision and the Environmental Law (No. 2872) and its regulations.

ER-14.02: Natural gas: NO_x ≤ 30 mg/Nm³, CO ≤ 30 mg/Nm³, NH₃ slip ≤ 5 mg/Nm³. LDO: NO_x ≤ 50 mg/Nm³, CO ≤ 50 mg/Nm³, SO₂ ≤ 10 mg/Nm³, dust ≤ 5 mg/Nm³, NH₃ slip ≤ 5 mg/Nm³. All dry, 15 % O₂, 50 % to 100 % GT load (ALP-OWN-ENV-001 section 1).

ER-14.03: Maximum 85 dB(A) at 1 m from any equipment in normal operation.

ER-14.04: Maximum 45 dB(A) LAeq night at the nearest sensitive receptor.

Parameter	Value	Condition	Status	Basis
CO	30 mg/Nm ³	dry, 15 % O ₂ , 50-100 % GT load	confirmed	ER-14.02
CO on LDO	50 mg/Nm ³	dry, 15 % O ₂ , 70-100 % GT load	confirmed	SRC-OWNR-0007, ER-14.02
Dust on LDO	5 mg/Nm ³	dry, 15 % O ₂	confirmed	SRC-OWNR-0007, ER-14.02
NH ₃ slip	5 mg/Nm ³	dry, 15 % O ₂	confirmed	ER-14.02
NO _x (as NO ₂)	30 mg/Nm ³	dry, 15 % O ₂ , 50-100 % GT load	confirmed	ER-14.02
NO _x on LDO	50 mg/Nm ³	dry, 15 % O ₂	confirmed	SRC-OWNR-0007, ER-14.02
SO ₂ on LDO	10 mg/Nm ³	dry, 15 % O ₂	confirmed	SRC-OWNR-0007, ER-14.02
Stack height	65 m	EIA decision (approved dispersion study)	confirmed	ER-06.06, SRC-OWNR-0007
Near-field noise	85 dB(A)	1 m from equipment	confirmed	ER-14.03
Noise at nearest receptor (day)	55 dB(A)	LAeq day	confirmed	SRC-OWNR-0007, ER-14.04
Noise at nearest receptor (night)	45 dB(A)	LAeq night	confirmed	ER-14.04

2.1 Where the design value is stricter than the EIA condition (for example the load range over which the NO_x and CO limits apply, or the night noise level defined in the Employer's Requirements), the design value is the one to be met and guaranteed; the EIA condition remains the legal limit.

3. Compliance matrix

Condition	Compliance measure	Evidence / document
EIA 1.1 Air: NO _x , CO, NH ₃ slip on natural gas (dry, 15 % O ₂ , 70-100 % load)	The GT combustion system, dry-low-NO _x , and the SCR with ammonia slip control are designed to the emission design parameters DP-CASE-0040 (NO _x), DP-CASE-0041 (CO) and DP-CASE-0042 (NH ₃ slip), which apply from 50 % GT load and therefore cover the 70-100 % load range of the EIA. IEC to confirm the values in the emission guarantee data; SCR catalyst volume and ammonia injection grid sized to the slip limit at end of catalyst life.	ALP-EPC-00000-HS-DBR-0001 (HSE design criteria); ALP-IEC-10MBA-PR-HMB-0001; emission test ALP-EPC-00000-CM-CMP-0006 / CM-TRP-0006. Status preliminary until the IEC emission guarantee is confirmed. [DP-CASE-0040, DP-CASE-0041, DP-CASE-0042, ER-14.02, SRC-OWNR-0007]
EIA 1.2 Air: NO _x , CO, NH ₃ slip, SO ₂ and dust on light diesel oil (LDO, at most the annual hours of the EIA)	LDO limits are held as DP-OWNR-0041 (NO _x), DP-EPCE-0009 (CO), DP-OWNR-0042 (SO ₂) and DP-OWNR-0043 (dust); the NH ₃ slip limit DP-CASE-0042 applies to both fuels; LDO operation is limited to the annual hours in DP-OWNR-0039 by the LDO management and reporting procedure; the LDO sulphur content DP-OWNR-0035 supports the SO ₂ limit.	ALP-EPC-00000-HS-DBR-0001; ALP-EPC-00000-PR-LST-0003 (chemical consumption, effluent and emission inventory); LDO operating hours counter reported per section 4. [DP-OWNR-0041, DP-OWNR-0042, DP-OWNR-0043, DP-OWNR-0039, DP-OWNR-0035, SRC-OWNR-0007, DP-EPCE-0009]
EIA 1.3 Air: stack height (result of the approved dispersion study)	The HRSG stack height is DP-CASE-0043 as approved in the EIA; the Owner's draft contract text differs and the EIA governs (TQ round of ALP-OWN-ENV-001). The stack foundation, structure and flue gas sampling levels are designed to this height. The dispersion confirmation with the final GT flue gas data is to be issued before the stack design is frozen.	ALP-EPC-00000-HS-STU-0002 (stack dispersion confirmation); ALP-EPC-10UHN-CV-FND-0002 (stack foundation pile layout). Status preliminary until the stack study is issued. [DP-CASE-0043, SRC-OWNR-0007]
EIA 1.4 Air: CEMS per EN 14181 with on-line reporting	One CEMS per HRSG stack, QAL1-certified analysers, QAL2 calibration and annual surveillance (AST) per the standard cited in reference REF-CONS-0084; data logger with the on-line emission data link to the authority (permit PMT-017) and to the Owner; CEMS scope covers NO _x , CO, NH ₃ , O ₂ and flow, plus SO ₂ and dust for LDO firing.	ALP-EPC-00CQA-IC-SPC-0001 (CEMS specification); ALP-EPC-00CQA-IC-TRP-0001 (QAL1, QAL2 and AST reports); ALP-EPC-00CQA-CM-CMP-0001; permit PMT-017. [REF-CONS-0084, PMT-017, SRC-OWNR-0007]
EIA 2.1 Water: cooling water temperature rise across the condenser at most 7 K	The circulating water flow (DEC-EPCE-0002) and the condenser are to keep the rise at or below DP-CASE-0015 in steady state in all ambient conditions (TQ-041). NOT MET in the cold cases: with the constant CW flow of the heat balance the rise is 7.20 K (MIN-NG-100) and 7.28 K (WIN-NG-100). IEC to raise the condenser design CW flow with the tube count and re-issue the cases (TQ-IEC-026); the hmb engine checks the rise of every case against DP-CASE-0015. Start-up and trip bypass transients up to 60 min are reported monthly (TQ-041).	ALP-EPC-00PAA-ME-CAL-0001 (CW sizing and hydraulic calculation); ALP-IEC-10MBA-PR-HMB-0001; ALP-EPC-00000-PR-HMB-0001 (plant heat balance, checks); ALP-EPC-00000-PR-STU-0003. Status open (TQ-IEC-026, see 5.1). [DP-CASE-0015, DEC-EPCE-0002, SRC-OWNR-0007, TQ-041, TQ-IEC-026, WIN-NG-100, MIN-NG-100]
EIA 2.2 Water: temperature excess at most 3 K at 100 m from the diffuser	The outfall diffuser is designed to keep the thermal plume excess at the edge of the mixing zone within DP-CASE-0016; this is performance guarantee PG-14 (must-rectify remedy, no tolerance) and is verified by the thermal plume survey of the EIA monitoring plan, summer conditions, base load. The seawater temperature basis is DP-OWNR-0012.	ALP-EPC-00000-PR-STU-0003 (CW thermal plume and outfall dispersion study); ALP-EPC-00UPA-CV-CVL-0001 (marine structures); guarantee PG-14 test procedure. Status preliminary until the plume study is issued. [DP-CASE-0016, PG-14, DP-OWNR-0012, SRC-OWNR-0007]
EIA 2.3 Water: residual oxidant (as Cl ₂) at the outfall at most 0.1 mg/l	Seawater electro-chlorination per DEC-EPCE-0015: continuous low-level dosing at the intake with pulse dosing; residual oxidant measured at the condenser outlet and at the outfall seal pit, dosing controlled on the seal-pit value with set-point 0.08 mg/l and alarm at the limit DP-CASE-0017; the compliance point is the end of the outfall pipe, sampled at the seal pit, no mixing zone credited (TQ-041); no dechlorination.	ALP-EPC-00PBA-PR-SDS-0001 (electro-chlorination SDD); ALP-EPC-00PBA-ME-CAL-0001; ALP-EPC-00PBA-ME-SPC-0001; TRO analysers at the condenser outlet and the seal pit (ALP-EPC-00PAA-IC-DSH-0001). Status preliminary until the SDD is issued. [DP-CASE-0017, DEC-EPCE-0015, SRC-OWNR-0007, TQ-041]
EIA 2.4 Water: process and sanitary effluents treated to the Water Pollution Control Regulation sea-discharge limits	Process effluents are collected by the process drains system and treated in the waste water treatment package with neutralisation by acid and caustic dosing (DEC-EPCE-0015); sanitary waste water is treated in a separate package; discharge quality is monitored and checked against the sea-discharge limits of the regulation in reference REF-CONS-0006; storm water discharge consent PMT-022.	ALP-EPC-00GNA-PR-SDS-0001 (waste water treatment SDD); ALP-EPC-00GUA-PR-SDS-0001 (sanitary); ALP-EPC-00GMA-PR-SDS-0001 (process drains); ALP-EPC-00000-PR-LST-0003; permits PMT-019, PMT-022. [REF-CONS-0006, DEC-EPCE-0015, PMT-019, PMT-022, SRC-OWNR-0007]

Condition	Compliance measure	Evidence / document
EIA 2.5 Water: SWRO brine not more than 5 % above ambient salinity at 50 m from the diffuser	DEC-EPCE-0015 returns the SWRO brine (00GBA10AP005 / 00GBA10AP006) to the CW outfall seal pit only while at least one CW pump runs; the SWRO trips on loss of all CW pumps and the demineralised water storage covers the outage. The brine is diluted at least 1:200 by one CW pump, so the salinity excess stays far below DP-EPCE-0012; ambient salinity is DP-OWNR-0015.	ALP-EPC-00GBA-PR-SDS-0001 (SWRO SDD); ALP-EPC-00GBA-ME-CAL-0001; ALP-EPC-00000-PR-STU-0003 (dilution); ALP-EPC-00PAA-IC-CNR-0001 (interlock). Status preliminary until the study is issued. [DEC-EPCE-0015, DEC-EPCE-0002, DP-OWNR-0015, 00GBA10AP005, SRC-OWNR-0007, DP-EPCE-0012]
EIA 3.1 Noise, operation: night and day limits at the nearest residential receptor	Plant noise is designed to the near-field limit DP-CASE-0044 and to the night limit at the nearest receptor DP-CASE-0045 (ER-14.03, ER-14.04); the noise study predicts the levels at the receptor with the equipment noise data and defines the attenuation (silencers, enclosures, cladding). The day limit of the EIA is DP-EPCE-0010.	ALP-EPC-00000-HS-STU-0001 (noise study); ALP-EPC-00000-HS-DBR-0001; noise survey ALP-EPC-00000-CM-TRP-0006. Status preliminary until the noise study is issued. [DP-CASE-0044, DP-CASE-0045, ER-14.03, ER-14.04, REF-CONS-0007, SRC-OWNR-0007, DP-EPCE-0010]
EIA 3.2 Noise, construction: noisy works (piling, blasting, steam blowing) only 07:00-19:00 weekdays and 08:00-17:00 Saturdays; steam blowing with 7 days' notice to the village and silencers	The construction method statements and the schedule restrict piling, blasting and steam blowing to the permitted hours; steam blowing is planned with temporary silencers and a 7-day notice issued through the Owner; the construction noise assessment report is a permit item (PMT-011).	ALP-EPC-00000-HS-PLN-0003 (construction environmental and social management plan); ALP-EPC-00000-CM-CMP-0001 (steam blowing procedure); permit PMT-011. [PMT-011, SRC-OWNR-0007]
EIA 4.1 Marine: no marine works (dredging, trenching, pipe laying) between 1 May and 30 September	The marine works (intake and outfall, CWP-06-CV-01) are planned inside the window of DP-EPCE-0014; the finish milestone CWP-06-CV-01W is constrained to finish no later than 30 April, so that the float of the marine works is measured against the closure and cannot move the works into the closed period. Permit PMT-006 (applied, 180 days lead time) is needed at the opening of the window.	ALP-EPC-00000-ME-PLN-0001 (marine operations plan); ALP-EPC-00000-CV-SPC-0005 (marine works specification); ALP-EPC-00UPA-CV-RPT-0001 (marine route survey); Level 3 schedule ALP-EPC-00000-PC-SCH-0001; permits PMT-006, PMT-007. [PMT-006, PMT-007, SRC-OWNR-0007, DP-EPCE-0014]
EIA 4.2 Marine: fish return system at the screens; intake velocity at most 0.15 m/s	The travelling band screens (equipment 00PAA10AT001 / 00PAA10AT002) are provided with a fish return trough and low-pressure screen wash; the intake structure is sized so that the through-screen velocity stays within the EIA value. The intake velocity limit is DP-EPCE-0011.	ALP-EPC-00PAA-PR-STU-0001 (intake hydraulic model study); ALP-EPC-00PAA-ME-CAL-0001; ALP-EPC-00PAA-PR-SDS-0001; ALP-EPC-00UPA-CV-CVL-0001. Status preliminary. [00PAA10AT001, SRC-OWNR-0007, DP-EPCE-0011]
EIA 4.3 Marine: monitoring programme during construction and 3 years of operation (Owner's obligation, Contractor provides data)	The Contractor provides the construction data (marine works logs, dredged or trenched volumes, turbidity records, vessel movements) and the operating data (CW flow, temperatures, chlorination dosing and residual, brine flow) to the Owner at the frequency defined in section 4.	ALP-EPC-00000-HS-PLN-0003; ALP-EPC-00000-ME-PLN-0001; data lists in section 4 of this document. [SRC-OWNR-0007]
EIA 5.1 Traffic: heavy vehicles avoid the village road 07:30-09:00 and 16:00-18:00	The abnormal load transport and delivery scheduling excludes these periods on the village road; heavy transport uses the route defined in the route survey, and the abnormal load permits are applied for with these restrictions (PMT-009). The construction traffic management is part of the CESMP.	ALP-EPC-00000-ME-RPT-0001 (abnormal load route survey and transport drawings); ALP-EPC-00000-HS-PLN-0003; permit PMT-009. [PMT-009, SRC-OWNR-0007]
EIA 5.2 Archaeological chance-find procedure applies to all excavations	The chance-find procedure is part of the CESMP and of every excavation permit-to-work: work stops, the find is protected and the notification is made through the Owner to the authority (permit item PMT-025); site staff receive the induction on the procedure.	ALP-EPC-00000-HS-PLN-0003; ALP-EPC-00000-HS-PLN-0002 (construction HSE plan); permit PMT-025. [PMT-025, SRC-OWNR-0007]
Environmental Law (No. 2872) via ER-14.01: waste management	Construction and operating waste is segregated, stored and disposed of through licensed carriers and facilities under the waste management plan; hazardous waste (used oil, spent catalyst, spent resins, chemical containers) has its own routes; the operating waste streams are quantified in the inventory.	ALP-EPC-00000-HS-PLN-0004 (waste management plan); ALP-EPC-00000-PR-LST-0003; ALP-EPC-00000-HS-PLN-0003. [ER-14.01, REF-CONS-0002]

4. Monitoring commitments

4.1 Emissions: continuous measurement of the CEMS parameters with on-line reporting to the authority and the Owner; LDO operating hours are logged per firing period and reported in the monthly environmental report; the emission tests ALP-EPC-00000-CM-CMP-0006 are performed during the Tests on Completion (ER-16.05).

4.2 Thermal plume: the plume survey of the EIA monitoring plan is carried out at base load in summer conditions (performance guarantee PG-14) and the seawater and outfall temperatures are recorded continuously during operation.

4.3 Cooling water and chemicals: CW flow, condenser temperature rise, chlorine dosing rate, residual chlorine at the outfall and SWRO brine flow are logged in the plant historian and reported monthly during the marine monitoring period (construction and 3 years of operation, Owner's obligation).

4.4 Noise: the operation noise survey at the nearest residential receptor is done during the Tests on Completion (report ALP-EPC-00000-CM-TRP-0006); during construction the noise records for piling and steam blowing are kept by the Contractor's HSE department and the 7-day steam blowing notice is filed.

4.5 Marine works: every marine works day is logged (activity, location, vessel) to demonstrate that no works were done between 1 May and 30 September.

4.6 Reporting: the Contractor submits a monthly environmental report to the Owner (conditions met, exceedances, complaints, open items of this matrix); an exceedance of an EIA limit is reported to the Owner within 24 hours with cause and corrective action.

5. Open items

5.1 Cooling water temperature rise (row 3.5): above the EIA limit in the cold cases with the heat-balance CW flow; TQ-IEC-026 asks IEC for the condenser design CW flow with the tube count and the re-issue of the heat balance cases; the CW pumps (00PAC10AP001/2) are re-rated with the answer.

5.2 Evidence documents not yet issued (status planned): noise study, stack dispersion confirmation, thermal plume study, CESMP and waste management plan; the rows depending on them stay preliminary until the documents are issued.

5.3 Resolved before this issue: the chlorine compliance point and the transients of the CW rise (TQ-041); the brine interlock and the seal-pit control of the residual oxidant (DEC-EPCE-0015); design parameters for CO on LDO, the day noise limit, the intake velocity, the brine salinity and the marine works window; the finish constraint of the marine works.

References

Codes and standards cited in this document (Codes, Standards and Regulations Register ALP-EPC-00000-GE-LST-0001; latest edition at contract date unless stated).

Code	Title
EIA Regulation	Environmental Impact Assessment Regulation (CED Yonetmeligi)
WPCR	Water Pollution Control Regulation (Su Kirliligi Kontrolu Yonetmeligi)
Noise Regulation	Environmental Noise Control Regulation (Cevresel Gurultunun Kontrolu Yonetmeligi)
EN 14181	Stationary source emissions - Quality assurance of automated measuring systems